

Website: <https://lewibs.com>
Github: <https://github.com/lewibs>

Benjamin Lewis
Phone: +1(919) 909-8267
Citizenship: American, Irish

Email: benjaminsl2000@gmail.com
Linkedin: <https://linkedin.com/in/lewibs>

Work Experience

Software Engineer	SMART Technologies	Aug 2024 - Present
--------------------------	---------------------------	---------------------------

- Recipient of the True North Award, recognizing exceptional contributions.
- Championed start to end development and managerial buy-in of multiple user facing features.
- Implemented a feature called "link anything" which allowed users to add to everything easily.
- Invented and led R&D for methods to detect and help struggling users.
- Integrated multiple AI features such as image background removal, LLM integrations, image sharpening, image retheming.
- Implemented right to left text support, unlocking a sales market which accounts for 10% of the global population.
- Worked on data design with the company quantitative analyst to help alleviate their workload.
- Integrated rotational math in the graphics engine to support rotated text.

Founder/Fullstack Software Engineer	Envision Reality	May 2023 - Aug 2024
--	-------------------------	----------------------------

- Envision Reality is a 3D user facing AI-assisted photorealistic interior design web-application.
- Created a 3D visualization engine that utilizes recent Gaussian Splatting research for AR/VR.
- Designed Lambda workers for integrations of long running third party APIs.
- Wrote WASM C scripts to implement faster graph traversal algorithms in the front end.
- Invented deep learning point cloud gaussian segmentation algorithms.

Front End Software Engineer	PowerN	Dec 2021 - Dec 2023
------------------------------------	---------------	----------------------------

- Architect for CPMS, a three.js based 4D computer vision engine for nuclear power plants.
- Converted wireframe designs into scalable well documented code and clean architecture.
- Designed scalable Django Python backend APIs and optimized front-end data flows.
- Implemented microservice for point-based compatibility checks for prefabricated components.
- Led an collaborative agile team of international developers and cross discipline engineers.

Technical Skills

-
- Languages: Javascript, TypeScript, Java, Python, C, C++, MATLAB, HTML/CSS
 - Frameworks: Pytorch, Three.js, Nest.js, Next.js, React, ReactNative, Potree, ifc.js, Flask
 - Tools: Linux, Docker, AWS, JWT, Mongo, Jest, WebAssembly (WASM), AWS-CDK, LIDAR, SQL

Education

North Carolina State University
Bachelor's Degree Computer Science

Raleigh, North Carolina
GPA 3.5

Open Source Contributions

Threejs - DragControls, added rotate mode - <https://github.com/mrdoob/three.js/pull/27689>
IFCjs - Raycaster improvements - https://github.com/ThatOpen/engine_components
Lumaapi - Updated to work with aws lambda <https://github.com/envisionreality/lumaapi-python>

Awards

True North Award

SMART Technologies, Nov 2024

Awarded to those who exemplify SMART's mission. My team and I received this for implementing right-to-left text support, unlocking a market representing 10% of the world's population.

Computer Science Honors

North Carolina State University, May 2023

Graduated with a minimum GPA of 3.5 while completing a challenging program of computer science undergraduate study, including an honors thesis with a supervising professor. Worked with Dr. Lina Battestilli to write a web application for enhancing security in child care.

1st place MATLAB Cody Competition

MATHWORKS NCSU, Sep 2021

Placed first place in the MATLAB Cody competition at North Carolina State University.

Dean's List

North Carolina State University, Dec 2023

Maintained 3.5 or better on 12 to 14 credit hours of coursework.

Projects

Ottery

<https://github.com/ottery-app>

- Childcare security system which awarded me the Computer science honors award.
- Designed and created a system for safely picking up and dropping off kids at daycares.
- Used by a local church to manage over 400 children.

Tech Stack: Nestjs, Javascript, React, Jest, Figma, Mongoose, MongoDB

Keyfile

<https://github.com/lewibs/keyfile>

- Created a DSL to simplify keyboard firmware with intuitive syntax for layers, macros, and colors.
- Developed transpiler to convert Keyfile syntax into C++ for QMK firmware.
- Is designed to allow direct compilation into binary firmware.

Tech Stack: C++, Linux, Docker, QMK

DuckTyper

<https://www.npmjs.com/package/ducktyper>

- Ducktyper is a runtime DTO validation tool to help with complexity involved with Forms and APIs.
- Created so that developers can create templates, then use those to validate input automatically.
- Heavily tested in Jest to allow for strong confidence in type checking correctness.

Tech Stack: Typescript, Jest, Node, npm

Guarded Components

<https://www.npmjs.com/package/guarded-components>

- Integrates with any JSX-compatible framework for dynamic authorization.
- Allows customizable contracts with conditions, state retrieval, and failure actions.
- Built with JavaScript and validated using Jest for production-ready stability.

Tech Stack: Javascript, React

6 Card Golf

https://github.com/lewibs/6_card_golf_ai

- Designed an AI agent to play 6 Card Golf, focusing on achieving the lowest game score.
- Taught models though, Q-learning, improving strategic gameplay with iterative learning.
- Decisions use statistical card values and pairing probabilities to minimize game score.

Tech Stack: Python, Pytorch